

In this paper, the authors explore the specialist capabilities of generalist foundation models like GPT-4 on medical challenge benchmarks in the absence of special training. The authors perform a systematic exploration of prompt engineering to boost performance. They find that general purpose prompt engineering methods with GPT-4 easily top prior leading results for medical question-answering datasets. The authors present Medprompt which greatly enhances GPT-4’s performance and achieves state-of-the-art results on all nine datasets in the MultiMedQA suite. The paper reports a 27% reduction in error rates and a score of over 90% on USMLE exam.

The authors present some key concepts like In-Context learning; where a foundation model is able to solve complex new tasks from a few demonstrations, Chain of Thought (CoT) prompting methodology that employs intermediate reasoning steps prior to introducing the sample answer, and Ensembling where outputs of multiple model runs are combined to arrive at a more robust answer. The datasets used in the experiments are from MedQA (containing MCQs in the style of medical licensing examination), MedMCQA (historic exam questions in the style of Indian medical school entrance exams), PubMedQA (questions requiring a yes, no, or maybe answer to biomedical research questions in reasoning required setting when given context provided from PubMed abstracts) and MMLU (multitask benchmark suite of 57 different datasets spanning domains across STEM, humanities, and social sciences).

The authors split the dataset into 80-20 training and testing data and used three major techniques, namely, Dynamic few-shot selection, self-generated chain of thought, and choice shuffle ensembling. For Dynamic few-shot, the authors selected k training examples that are semantically similar using a k-NN clustering. In self-generated chain of thought, the model is asked to provide intermediate reasoning steps. To avoid hallucinated or incorrect reasoning chain, if the final answer does not match the ground truth label, the entire sample is discarded. GPT-4 may tend to favor some options in MCQs over others. To mitigate this, the authors shuffle choices and check for consistency of the answer. Medprompt combines intelligent few-shot exemplar selection, self-generated chain of thought steps, and a majority vote ensemble.

Their experimental results show that GPT-4 with Medprompt achieved an average performance of 90.6% on the eyes-on data, and an average performance of 91.3% on the eyes-off data, suggesting that the prompt engineering process likely did not lead to overfitting on MultiMedQA datasets. The data also shows that the chain-of-thought steps contributed the most to performance (3.4% increase), followed by few-shot prompting and choice shuffle ensembling (2.2% increase each). However, the authors do note that these techniques are not statistically independent and the order in which each contribution is tested matters and is subjective.

The authors note that the techniques used in this study are not specific to medical question-answering and can be used in other applications. To validate this claim, they tested the final Medprompt methodology on six additional, diverse datasets from the MMLU benchmark suite covering electrical engineering, machine learning, philosophy, professional accounting, professional law and criminal psychology, as well as datasets for NCLEX (Registered Nurse exam) style questions. Their results show that their approach managed to score 89%+ across all sets.

In my opinion, the research is fascinating but AI lacks what you can call the “human” or “trust” factor. An AI lawyer, judge or medical practitioner may be able to score high on the theoretical exams but it would lack empathy and reasoning beyond a certain threshold over textbook material. The authors also note that as foundation models are trained on massive, internet-scale datasets, strong performance on benchmark problems may be due to memorization or leakage effects, where direct test samples have previously been observed by the model during training. These AI models can and should be used to aid humans but solely depending on them still seems to be unreasonable. The study also indicates that it is the time to rethinking these standardized tests for humans as well. Lastly, I am glad that its not just my job which is at risk of being replaced by AI ☺.